

ALAN MAN

SOFTWARE ENGINEER · FULL STACK DEVELOPER

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years of experience building enterprise web and mobile applications for hospitality and insurance. Experienced with React, TypeScript, Python/FastAPI, Java Spring Boot, SQL, Microsoft Azure and React Native. Build REST APIs, secure workflows, offline-capable applications and cloud deployments. Contributed to a hotel case management platform that helped shift most customer enquiry responses from 1–2 days to the same day. Authorized to work in Canada.

TECHNICAL SKILLS

Languages	TypeScript · JavaScript · Python · Java · SQL
Frontend & Mobile	React · React Native · Redux Toolkit · Material UI · Vite · Angular · HTML · CSS
Backend	FastAPI · Spring Boot · NestJS · Azure Functions · REST APIs
Databases & Data	Azure SQL Database · SQL Server · IBM DB2 · MongoDB · Realm.js · SQL query optimization · SqlPackage CLI
Cloud & DevOps	Azure App Service · Azure Storage Queue · Azure DevOps YAML Pipelines · Azure Key Vault · Managed Identity · Deployment slots · Microsoft Graph API · Jenkins
Security & Testing	Microsoft Entra ID · MSAL · JWT · RBAC · Pydantic · Pytest · Vitest · Integration testing · UAT

PROFESSIONAL EXPERIENCE

May 2025 – Present

System Developer

Eastech Systems Limited · Remote

Core developer of Mercury, an AI-powered enterprise hotel messaging and case management platform for an international hospitality organization.

Key Achievements & Contributions

- **Established an enterprise-scale hotel messaging and case management platform**, processing 62,000+ emails and 20,000+ cases during Q1–Q2 2026.
- **Drove measurable operational improvements**, reducing average email processing time by **74% (10.8 to 2.8 hours)** and case completion time by **72% (89 to 25 hours)** through centralized workflows and AI-assisted automation.
- **Developed full-stack case management workflows** using React, TypeScript, Python, FastAPI, and

Azure SQL, including case assignment, bulk operations, status tracking, and audit history.

- **Integrated AI-assisted workflows** for email analysis, classification, tagging, summarization, urgency assessment, recommended actions, and draft responses.
- **Designed and implemented granular RBAC** with feature-level permissions, user-role assignments, and administrative tools for access management.
- **Implemented secure enterprise authentication and integrations** using Microsoft Entra ID, MSAL, JWT, and Microsoft Graph API.
- **Optimized SQL queries and backend performance**, engineering asynchronous workflows using Azure Functions and Azure Storage Queue.
- **Engineered and managed Azure cloud infrastructure and production deployments** using Azure App Service, Azure DevOps CI/CD, Key Vault, managed identities, and deployment slots.
- **Developed operational dashboards** for monitoring case activity, agent workloads, and workflow performance.

Tech Stack: React, TypeScript, Python, FastAPI, Azure SQL, Microsoft Graph API, Azure Functions, Azure DevOps.

March 2021 – January 2025

Analyst Programmer

Arch-Partnership Limited

Developed enterprise web and mobile insurance applications for Prudential, supporting high-volume agency and brokerage operations, digital policy submissions, offline workflows, remote signing, and payment processing.

Key Achievements & Contributions

- **Established enterprise insurance application workflows** serving 300+ agents and brokers, processing an estimated 100,000+ insurance applications over multiple years.
- **Accelerated insurance application processing from approximately 1 week to 1–3 days** through workflow improvements, automated validation, electronic signatures, and streamlined submissions.
- **Streamlined end-to-end policy applications** by developing Angular applications integrated with Prudential's internal systems through REST APIs, data mapping, and business rules.
- **Engineered offline-first mobile applications** using React Native and Realm.js, implementing synchronization validation, conflict detection, and data recovery.
- **Enabled secure remote insurance signing** through VoIP-based signing workflows and facial recognition identity verification.
- **Digitized document and payment processing** through multi-party e-signatures, offline PDF generation, payment gateway integration, and transaction verification.
- **Automated insurance calculations and eligibility checks** using Java Spring Boot and NestJS services integrated with external insurer APIs.
- **Delivered secure enterprise applications** using IBM DB2, MongoDB, JWT authentication, Jenkins, and Azure Pipelines.

Tech Stack: Angular, React Native, Realm.js, Java, Spring Boot, NestJS, REST APIs, IBM DB2, MongoDB.

July 2020 – September 2020

Software Engineer (Part-time)

Woow Moment Limited

Developed cross-platform mobile applications, backend APIs, and location-based services for Ebatte, a

connected power bank platform, and Run2Gather, a running community application.

Key Achievements & Contributions

- **Enabled mobile-to-device connectivity** for Ebatte's power bank platform by developing React Native native-module bridges to integrate JavaScript application logic with native mobile functionality and Bluetooth beacon communication.
- **Engineered RESTful backend APIs** using Node.js and Express.js to facilitate communication between mobile applications, backend services, and connected power bank devices.
- **Improved frontend maintainability and usability** by developing reusable, responsive React.js components for device integration and management workflows.
- **Integrated GPS-based geolocation and Google Maps SDK** into Run2Gather, enabling location-based running activities, route visualization, and interactive mapping.
- **Developed community-focused mobile features** for Run2Gather, supporting running event organization and activity tracking.

Tech Stack: React Native, React.js, JavaScript, Node.js, Express.js, REST APIs, Bluetooth Beacons, Google Maps SDK, GPS, Geolocation Services.

June 2020 – January 2021

Student Helper, Path Advisor

Hong Kong University of Science and Technology (HKUST)

Contributed to **Path Advisor**, HKUST's campus mapping and indoor navigation platform, integrating location-based services, indoor positioning technologies, and interactive navigation features.

Key Achievements & Contributions

- **Enhanced campus navigation and route-finding functionality** by developing and maintaining React.js features for interactive campus maps, building navigation, and route search.
- **Advanced indoor localization in GPS-denied environments** by contributing to a crowdsourced Indoor Positioning System (IPS) leveraging Wi-Fi Received Signal Strength Indicator (RSSI), Bluetooth Low Energy (BLE) beacon localization, and multi-sensor fusion for indoor positioning and floor-level identification.
- **Developed RESTful backend services** using Node.js, Express.js, and MongoDB to support location data retrieval, route search, and campus mapping, integrating Realm DB for application-side geolocation data.
- **Improved indoor mapping coverage and navigation accuracy** by collecting wireless signal measurements, capturing Ricoh 360° imagery, and maintaining floor plans, building layouts, and pedestrian routes.
- **Improved application reliability and usability** through feature testing, route validation, defect identification, and troubleshooting across frontend and backend components.

Tech Stack: React.js, JavaScript, Node.js, Express.js, MongoDB, Realm DB, Indoor Positioning Systems (IPS), Wi-Fi RSSI, BLE Beacons, Sensor Fusion, Geolocation, REST APIs.

September 2018 – August 2019

IT Student Helper

Audit Commission, HKSAR Government

Provided IT support and infrastructure administration for the Audit Commission, covering Windows Server environments, identity management, internal web application hosting, and data backup.

Key Achievements & Contributions

- **Maintained reliable daily IT operations** by administering Windows Server environments and troubleshooting workstation, network, email, and user access issues for government staff.
- **Supported secure user access and centralized identity management** through Active Directory administration, account provisioning, and access permission management.
- **Supported internal web application hosting and infrastructure availability** by configuring and maintaining Apache Tomcat, Microsoft IIS, and IBM WebSphere Application Server environments, alongside VMware-based virtual server setup and administration.
- **Strengthened data protection and business continuity** by supporting NAS-based backup solutions and Windows Server replication infrastructure, helping maintain data availability and recovery capabilities for departmental systems.

Tech Stack: Windows Server, Active Directory, Apache Tomcat, Microsoft IIS, IBM WebSphere Application Server, VMware, NAS Backup, Windows Server Replication.

OTHER EXPERIENCE

October 2026

BuildYYC Buildathon

Team Participant · Calgary, AB

- Placed 4th overall at the BuildYYC Buildathon in Calgary with a 5-person team, taking PlateWise from idea to final solution.
- Identified the ideal customer profile (ICP) and ran customer validation with 15–20 potential customers to confirm the problem and demand.
- Assessed the market opportunity and analyzed competitors to define clear points of differentiation.
- Shaped the final solution and established the business model, go-to-market plan and product roadmap.

EDUCATION

Bachelor of Engineering, Computer Engineering — 2021
Hong Kong University of Science and Technology

Associate in Engineering — 2015
Hong Kong Community College